Project 5: Fast Fourier Transform Accelerator

Introduction

The Fast Fourier Transform (FFT) is an algorithm to efficiently compute the Fourier Transform with a computational complexity in $\mathcal{O}(NlogN)$ rather than $\mathcal{O}(N^2)$ if the formal definition is used. The FFT algorithm finds applications in almost every field where spectral analysis is needed: in biomedical and audio signal processing, in power quality and vibration analysis but also in wireless communication and radar technologies . Even though the FFT algorithm is already efficient, it remains computationally intensive as it requires $\log(N)$ stages of computation. Dedicated FFT hardware accelerators are therefore common, with challenges to make them both energy and resource efficient while maximizing their throughput.

Project Description

In this project, you will design and implement a hardware FFT accelerator using SystemVerilog. You will need to understand the FFT algorithm and the different hardware blocks required to implement it. You will explore accelerator as well as datapath architectures and evaluate their impact on resource usage and performance. You will implement the datapath, control logic and pipeline to improve these metrics. After simulation and verification, you will synthesize the design on an FPGA and measure its throughput, latency, and area.

Expected Results

You will produce a functional and verified FFT hardware module optimized with respect to your target performance metric. The design should demonstrate significantly higher performance and energy efficiency than software implementations, showcasing how custom hardware accelerators enhance the speed and scalability of spectral analysis.

Useful References

1. Documentary (26 minutes) on the FFT algorithm and history : Video
2. Vercruysse A.: A Tutorial-style Single-cycle Fast Fourier Transform Processor (source code also available)
3. Kaya Z : An Area-Efficient Normal Input/Output Ordered Memory-Based FFT Using an SC Kernel (Available on the course website)
4. Garrido M. : A Survey on Pipelined FFT Hardware Architectures